



Inspiring Excellence

Network Layer: Subnetting

Lecture 8 | CSE421 – Computer Networks

Department of Computer Science and Engineering
School of Data & Science

Objectives

- IPv4 Exhaustion
- Solution to the depletion of IPv4
- Types of Subnetting
- Examples

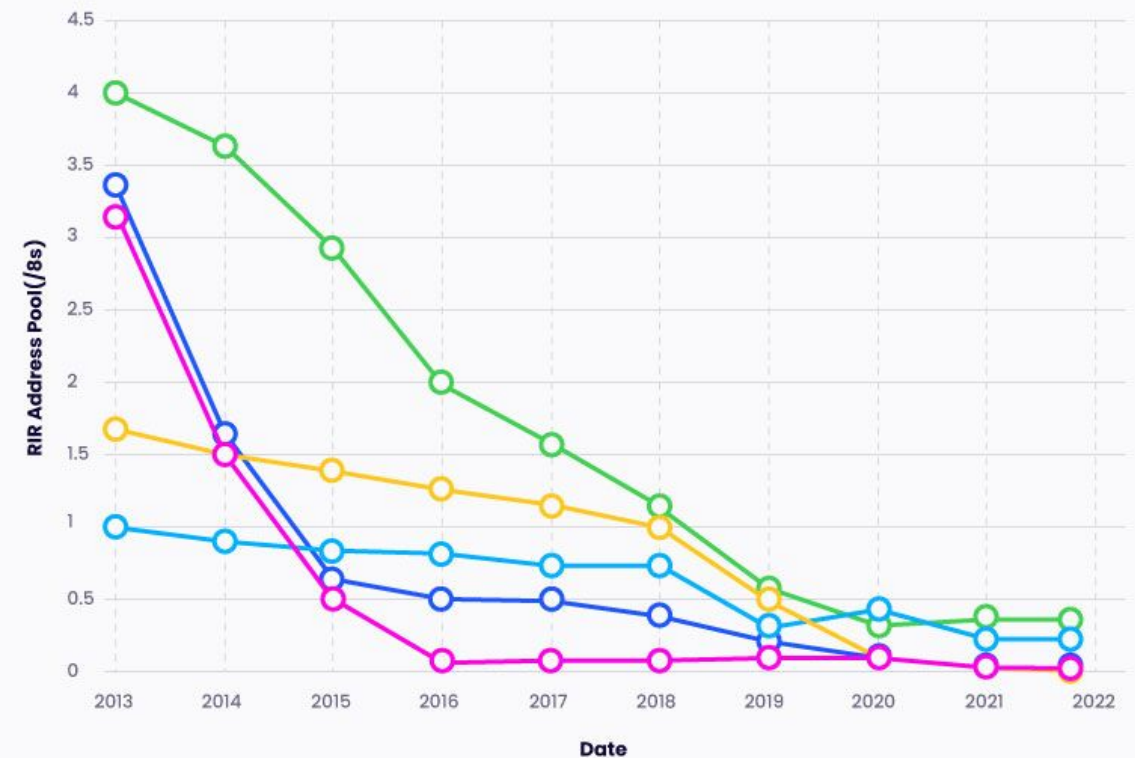
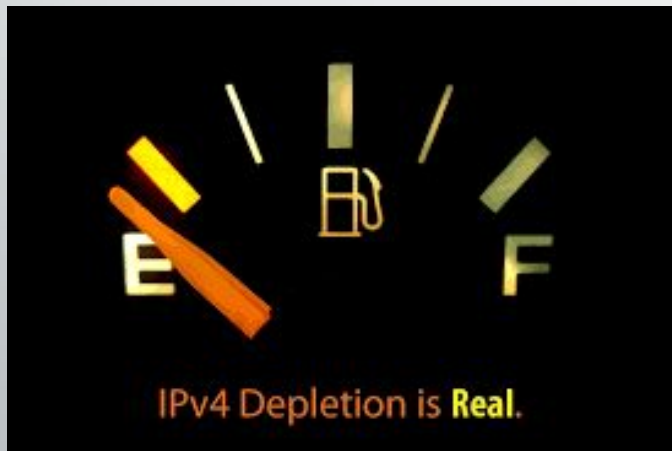
IPv4 Address Exhaustion

This report generated at 10-Jan-2016 08:20 UTC.

IANA Unallocated Address Pool Exhaustion:
03-Feb-2011

Projected RIR Address Pool Exhaustion Dates:

RIR	Projected Exhaustion Date	Remaining Addresses in RIR Pool (/8s)
APNIC:	19-Apr-2011 (actual)	0.6284
RIPE NCC:	14-Sep-2012 (actual)	0.9520
LACNIC:	10-Jun-2014 (actual)	0.1140
ARIN:	24 Sep-2015 (actual)	
AFRINIC:	12-Aug-2018	1.8246



ARIN RIPE NCC APNIC LACNIC AFRINIC

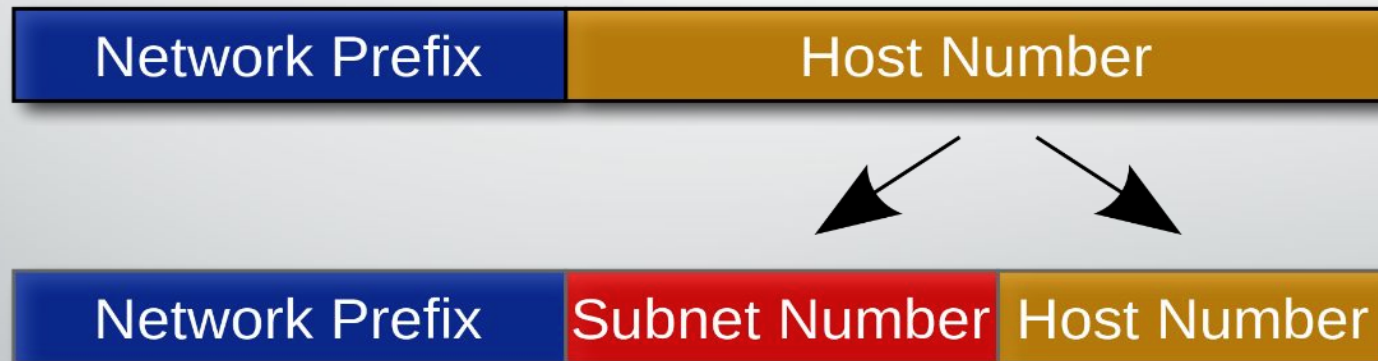
Solutions

- **Long term:**
 - Change to IP version 6.
 - Plenty of addresses using a different scheme
- **Short term:**
 - Use **Subnetting** to avoid wasting addresses
 - Use **private addresses** locally and **NAT** for internet access – lets many host share a few public addresses

Private IP address space	
From	To
10.0.0.0	10.255.255.255
172.16.0.0	172.31.255.255
192.168.0.0	192.168.255.255

Subnetting

- Partition a single physical network into more than one smaller logical sub-networks (subnets).
- **Borrow bits** from **the IP address's host part**
- Use these bits to create a number of smaller sub-networks inside the original network.



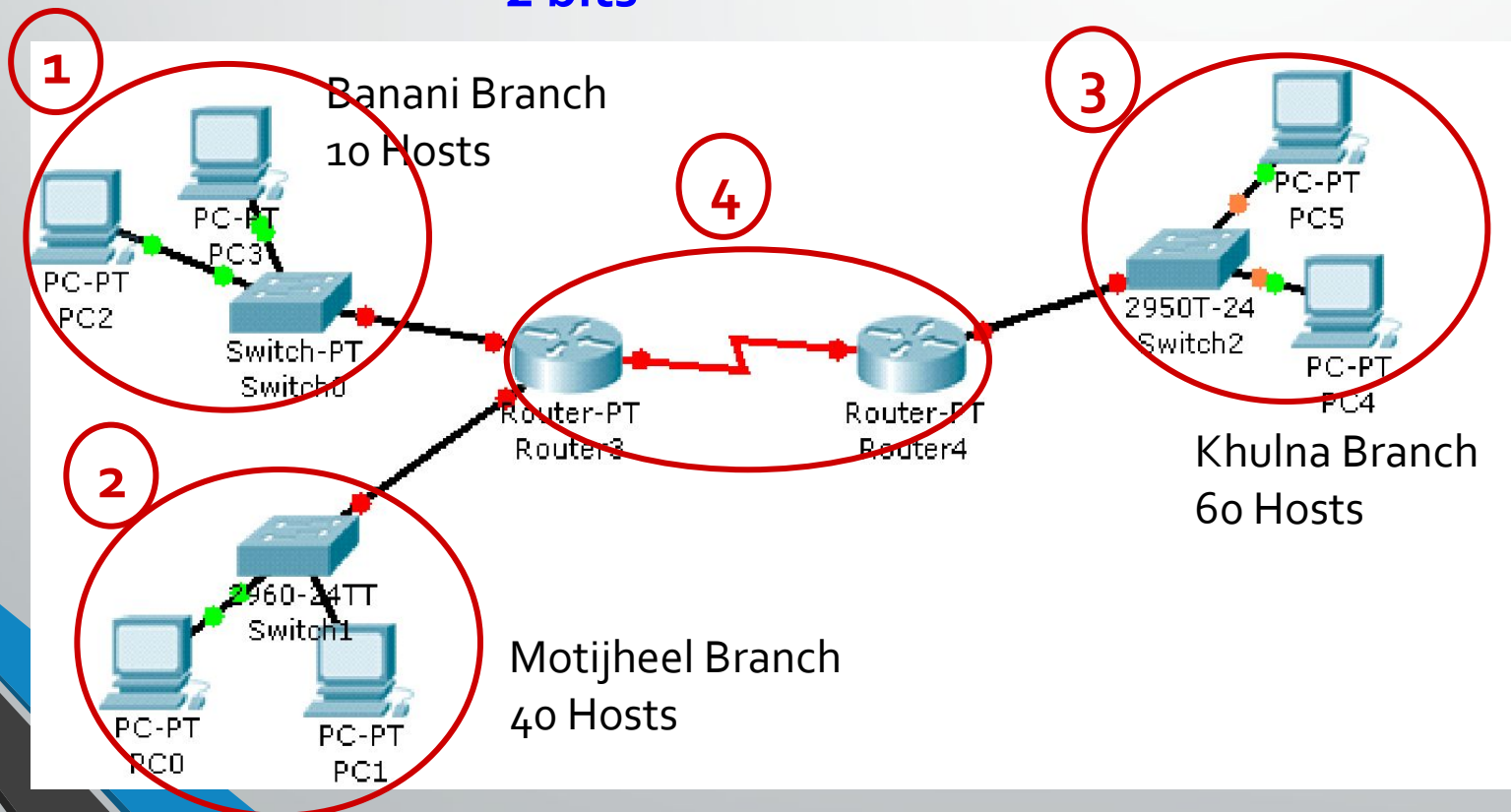
Subnetting

- Two methods of subnetting
 - Fixed Length Subnet Masking
 - Variable Length Masking

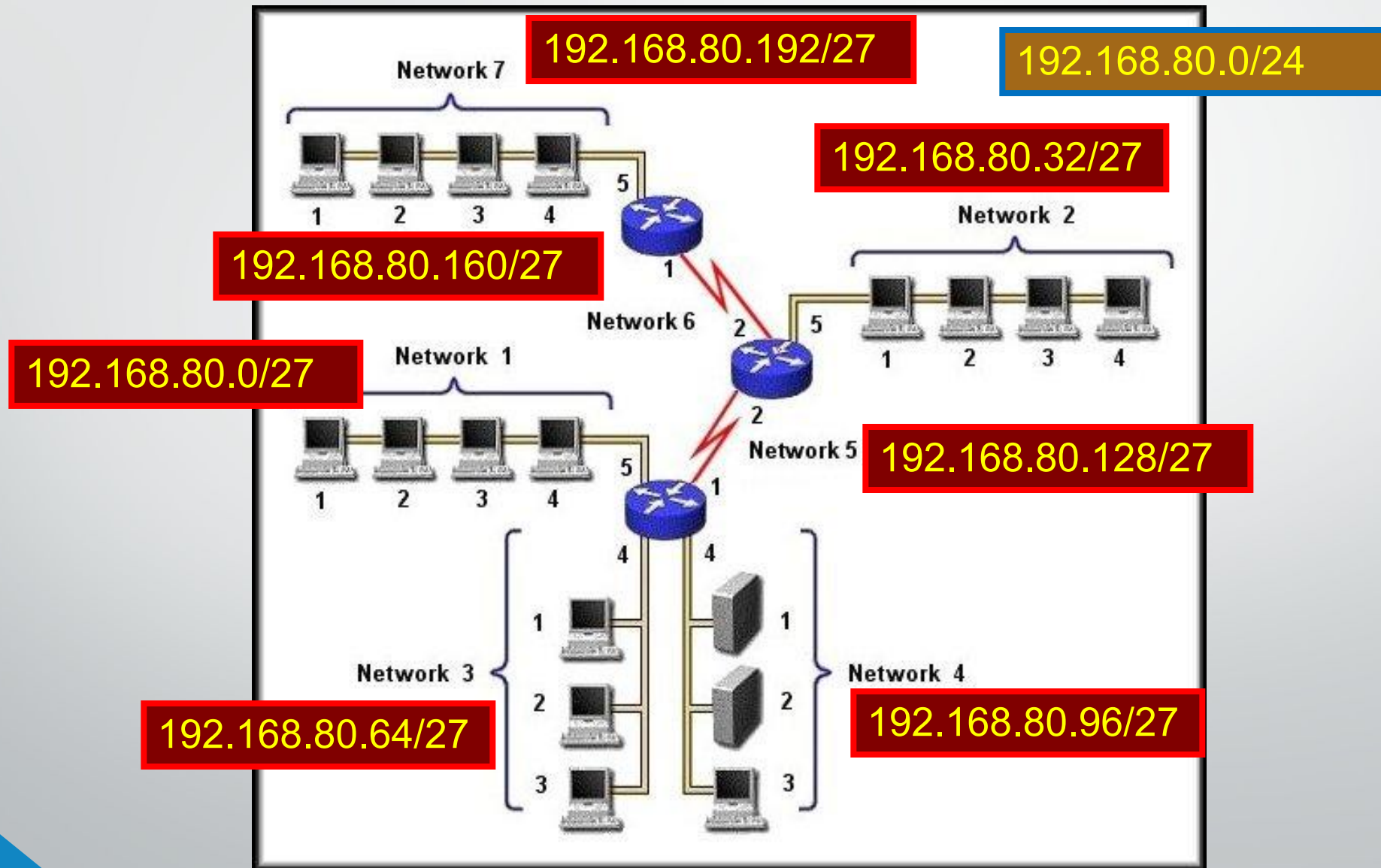
Fixed length subnetting

- How many network addresses do you need for the organization?
- How many network addresses do you have?
- How many bits do you have for hosts? **8 bits**
- How many host bits do you need to borrow to create the number of sub networks that you need? **2 bits**

200.20.20.0/24



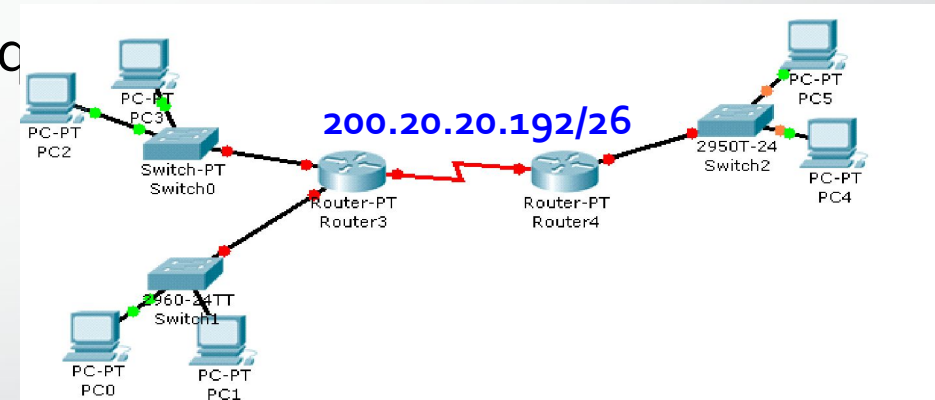
Fixed Length Subnetting: Example



Lots of Waste!

- **Problem of Fixed Subnetting:**

- Waste IPv4 addresses.
- For example : Router3 –Router2 Network requires **62** addresses. **60** are wasted.
- How many is available? How many wasted?
- If you are using private addresses, then you may not be bothered.
- Waste of public addresses does matter.



- **Solutions:**

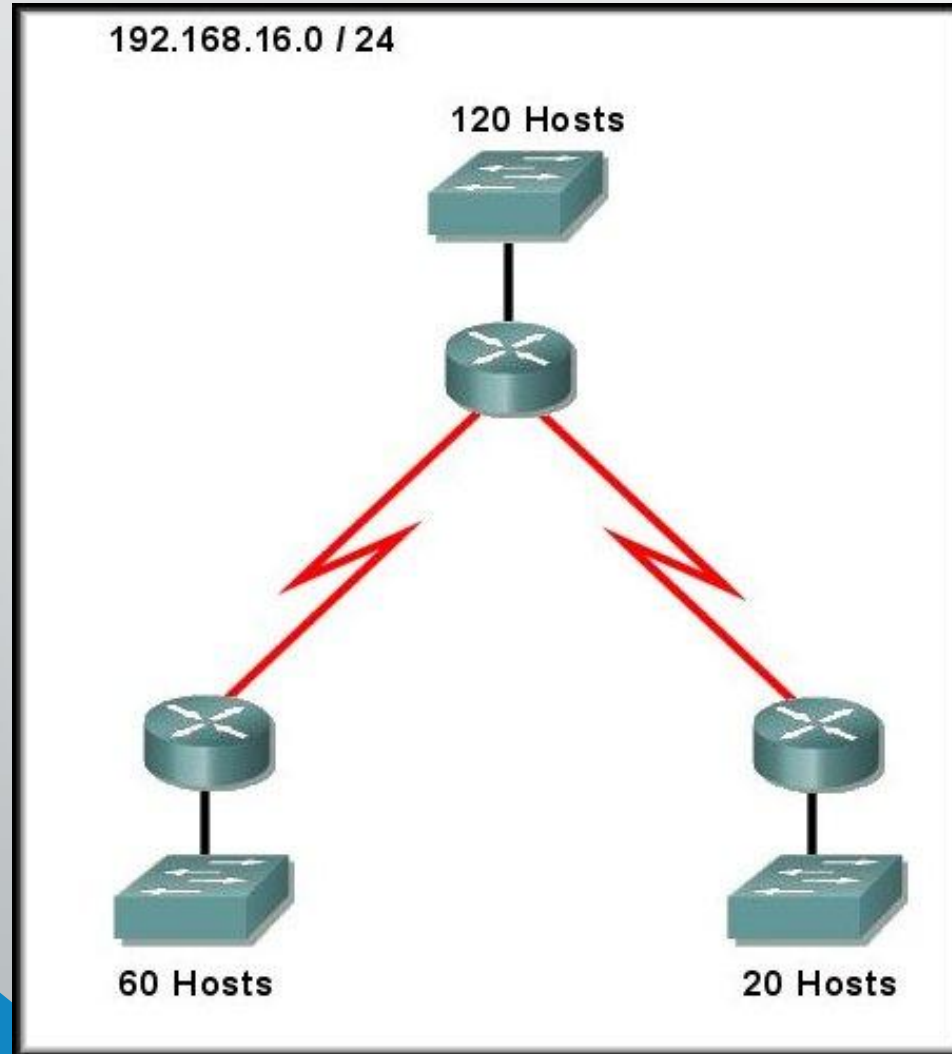
- **Variable Length Subnet Masking (VLSM)**

- Create subnets as per specific host requirements.

VLSM

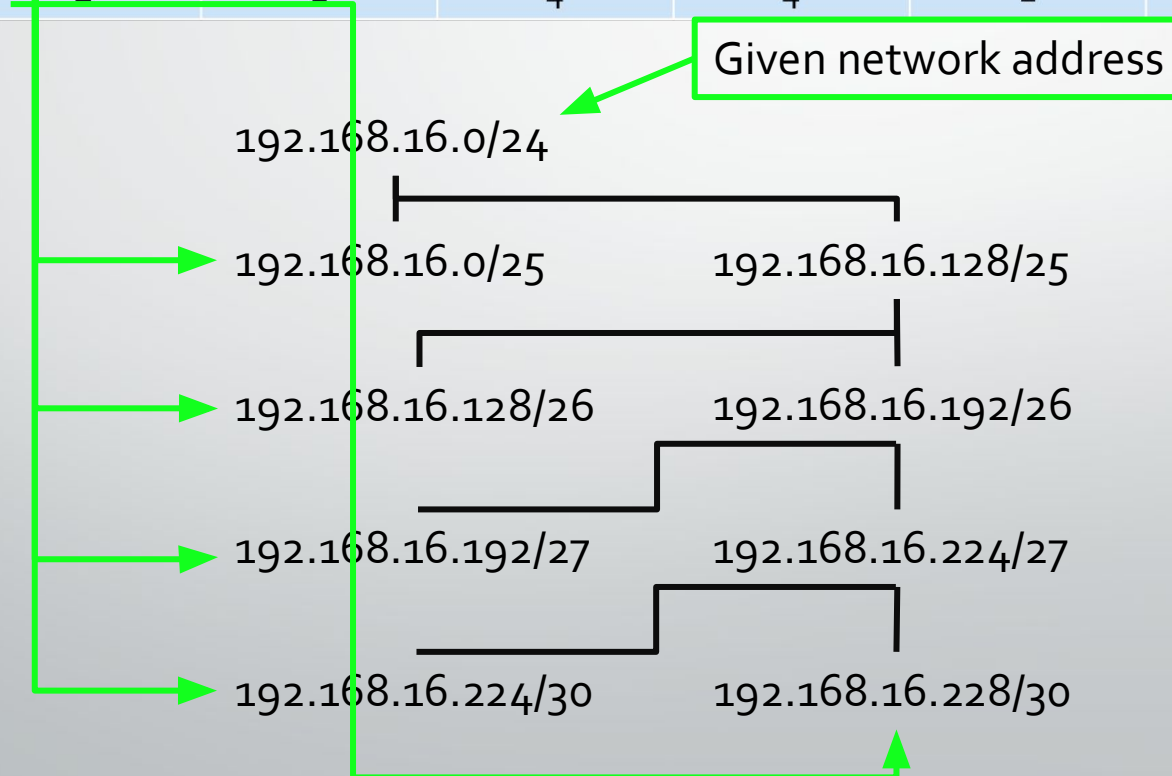
- Also known as “**Variable Length Subnet Masking**”
 - Assign a block of IP satisfying only that particular LAN.
 - **200 hosts?** Assign a block of size 256 IPs!
 - **1000 hosts?** Assign a block of size 1024 IPs!
 - **2 Hosts?** Assign a block of size 2 IPs!
- Always satisfy the requirements of your biggest LAN and then work your way down to the smallest LAN.

VLSM Example 1



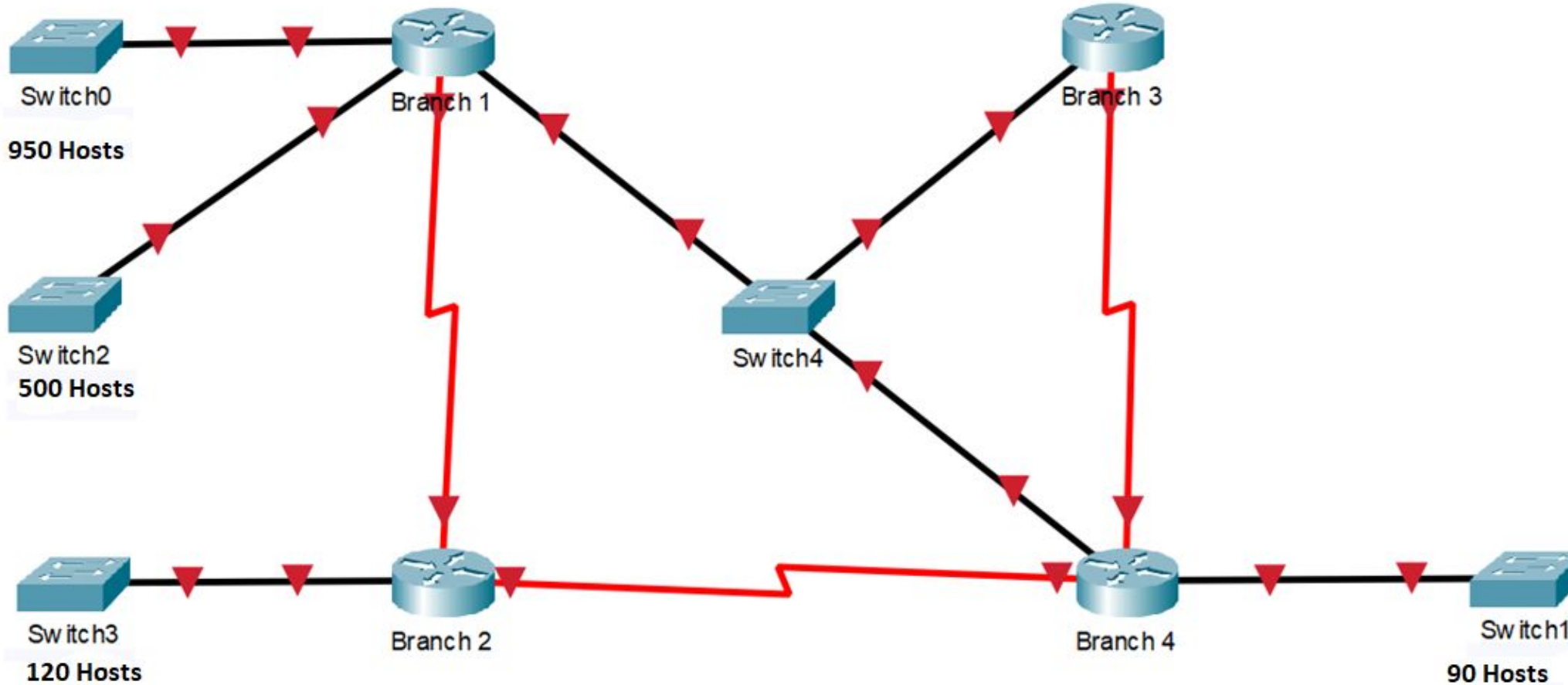
VLSM Example 1

Host Name	Host requirement	+2	IP Block size	Host bits required	Network bits required
A	120	122	128	7	25
B	60	62	64	6	26
C	20	22	32	5	27
D	2	4	4	2	30
E	2	4	4	2	30



VLSM Example 2

Given Network Address : 13.65.136.0/21



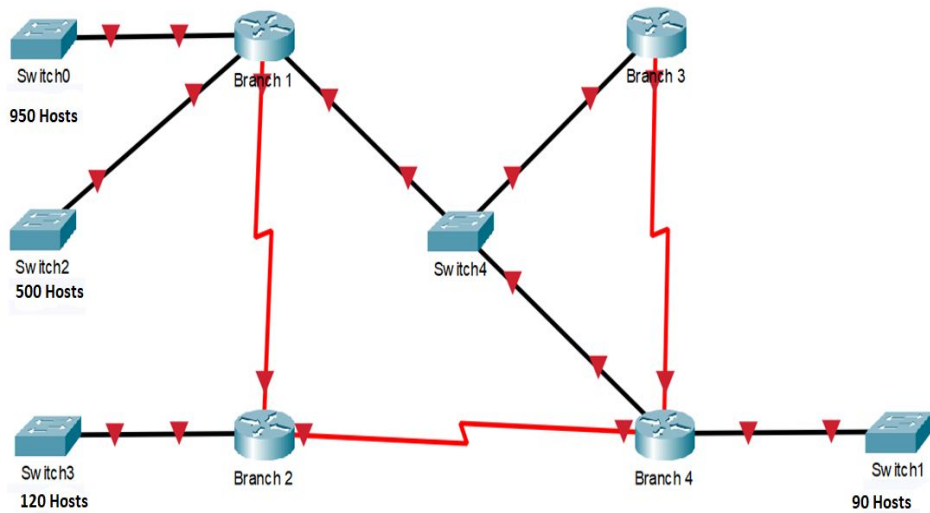
4 LANs

3 WANs

1 Switched Network

VLSM Example 2

Given Network Address : 13.65.136.0/21



The End